CS5361/6361 Machine Learning

Spring 2026

Arrays and Numpy - Quiz 1

Write the output produced by each of the *print* statements in the code below.

```
import numpy as np
a = np.array([-1,3,2,0])
for i in range(a.shape[0]):
    print(i, a[i], a[a[i]])



a = np.array([[4,1,0,5],[0,3,1,3],[2,1,0,3]])
print(a)



print(a.shape)



print(len(a.shape))



print(a[1])



print(a[-1])



print(a[0,2])



print(a[:,1])



print(a + 1)



print(a.reshape(-1))



for i in range(3):
    print(a[i,i])
```

```
a = np.array([[4,1,0,5],[0,3,1,3],[2,1,0,3]])
print(np.sum(a))



print(np.sum(a,axis=0))



print(np.sum(a,axis=1))
```